

Head Movement During Five Activities of Daily Living

A Single Participant Smartphone IMU Feasibility Study

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Abstract

This study tested a simple workflow for recording head movement during five activities of daily living using a smartphone IMU fixed to the forehead. One healthy adult completed three trials of seated laptop work, standing laptop work, standing phone use, eating, and level walking, together with one active calibration recording. Each task trial was resampled to 50 Hz, referenced to its own 10 s neutral hold, and analysed over a fixed 180 s window. Mean flexion-extension excursion ranged from 9.7 deg during standing laptop work to 19.5 deg during standing phone use; the corresponding coefficients of variation ranged from 21% to 46%. Quiet-hold noise averaged 1.12 deg, while neutral phone inclination spanned 20.5 deg across the fifteen recordings, showing that stable sampling did not produce strong between-trial repeatability. The project demonstrates a consistent acquisition and processing workflow, but it does not validate cervical range of motion because one head sensor cannot separate neck motion from trunk motion and no reference instrument was used.

Introduction

Daily activities use different portions of available cervical movement [1]. This is relevant to assistive neck devices because functional head movement, rather than maximum range alone, affects tasks such as screen use, eating, conversation, and walking [2]. Inertial sensors can estimate head or cervical orientation under controlled conditions [3,4], but a sensor attached only to the head measures head orientation in space and cannot separate cervical motion from trunk motion.

The objective was to collect head-orientation data during five simple activities and process every recording with the same MATLAB workflow. This was an independent single-participant feasibility study, not a clinical study or a source of population reference values.

Methodology

A smartphone was centred on the forehead and held by a cap strap passed between the phone and its cover. The author completed one 60 s active calibration and three recordings of each task. Walking consisted of repeated straight indoor laps separated by whole-body turns.

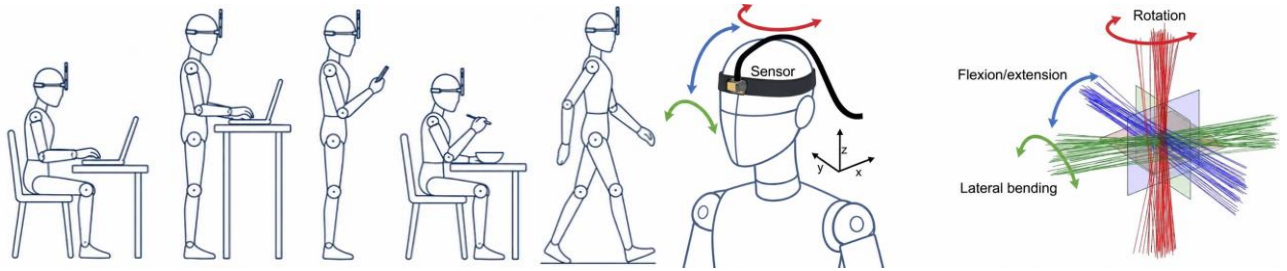


Figure 1 Recorded task positions. The seated laptop was 54 cm away with the screen top 18 cm below eye level. The standing laptop was 62 cm away with the screen top 12 cm above eye level. The forehead phone was held by a strap between the phone and cover. Schematic not to scale.

(AI disclosure: The mannequin illustrations were generated using OpenAI's image-generation tool and manually arranged and annotated by the author.)

Gyroscope and acceleration data were placed on a uniform 50 Hz time grid. The quietest complete 10 s block in the first 30 s defined the reference orientation for each recording. Gravity was isolated with a zero-phase second-order Butterworth low-pass at a 0.5 Hz design cutoff, a value evaluated for gravity separation in accelerometry [5]. Flexion-extension and lateral bending were estimated from the gravity direction in a trial-specific head frame. Axial rotation rate was bias-corrected against the neutral hold, filtered with a zero-phase second-order Butterworth low-pass at a 5 Hz design cutoff, in the same range used in comparable head-kinematics work during activities of daily living [6], then integrated. A fixed 180 s window began 2 s after the neutral hold, and the integrated signal was linearly detrended within that window. For walking, samples within 1 s of any axial rotation rate above 25 deg/s were set to NaN, following angular-velocity turn detection used for wearable IMUs [7]. Walking axial rotation was not reported.

Excursion was the 97.5th percentile minus the 2.5th percentile. Trial-level values were summarized using the mean, sample standard deviation, sample variance, and coefficient of variation, $CV = 100 \times SD / \text{mean}$. All repeatability statistics are descriptive because each task had three trials [8]. Flexion-extension is the primary outcome. The active calibration measured

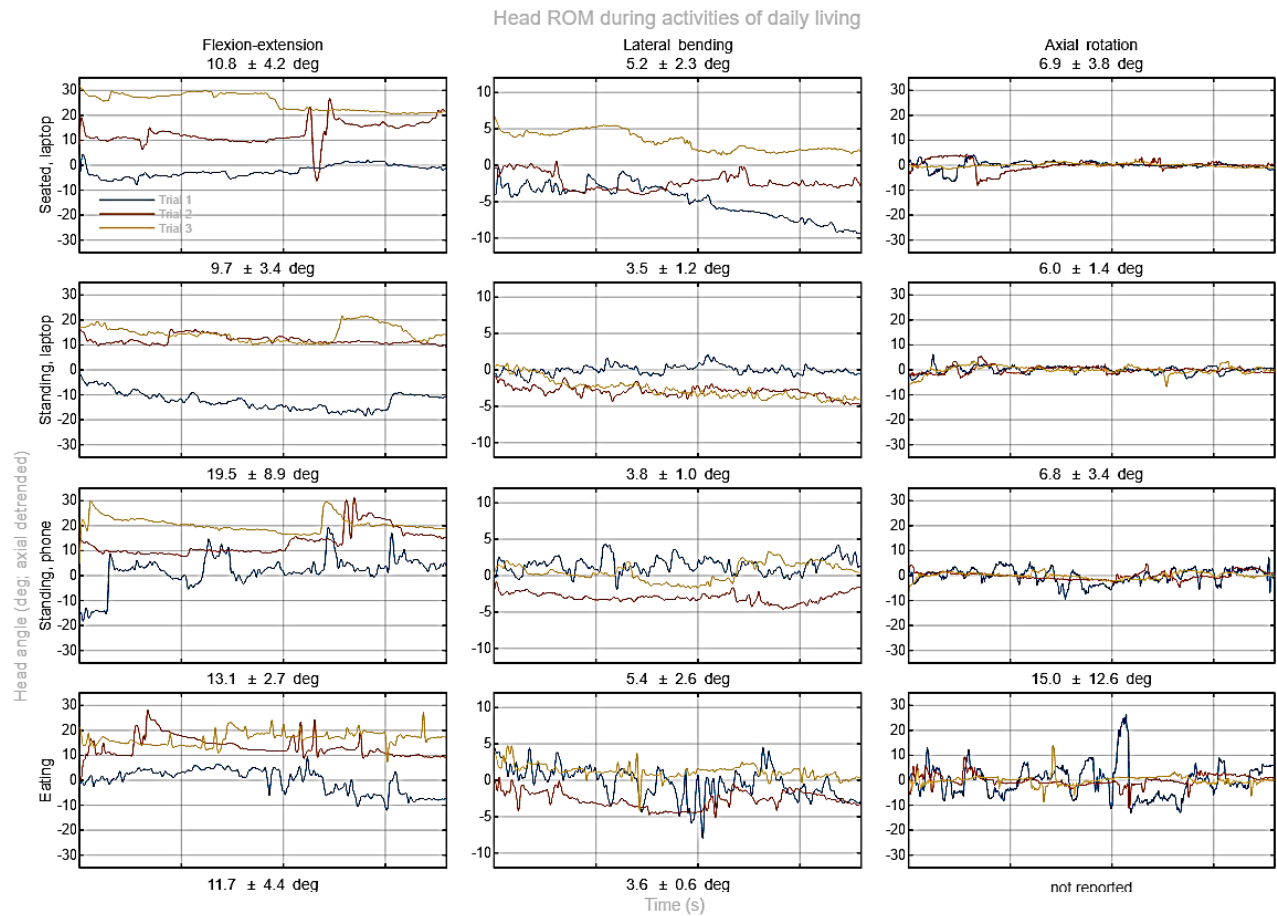
sagittal range only, so the percentage of available range can be computed for that plane alone. Lateral bending and axial rotation are reported as context.

Results

All fifteen task recordings were processed. The measured sampling rate averaged 49.92 Hz and quiet-hold noise averaged 1.12 deg. The active calibration produced a sagittal range of 119.6 deg. Mean flexion-extension excursion ranged from 9.7 to 19.5 deg, equal to 8%-16% of the calibrated sagittal range. Walking retained 77% of samples after turn exclusion.

Task	Flexion-extension mean +/- SD (deg)	Variance (deg ²)	Coefficient of Variation (%)	Lateral bending (deg)	Axial rotation (deg)	Active range (%)
Seated laptop	10.8 +/- 4.2	17.7	39	5.2	6.9	10
Standing laptop	9.7 +/- 3.4	11.7	35	3.5	6.0	9
Standing phone	19.5 +/- 8.9	79.6	46	3.8	6.8	16
Eating	13.1 +/- 2.7	7.5	21	5.4	15.0	11
Walking	11.7 +/- 4.4	19.1	37	3.6	NR	10

Table 1 Head excursion and flexion extension repeatability across three trials per task



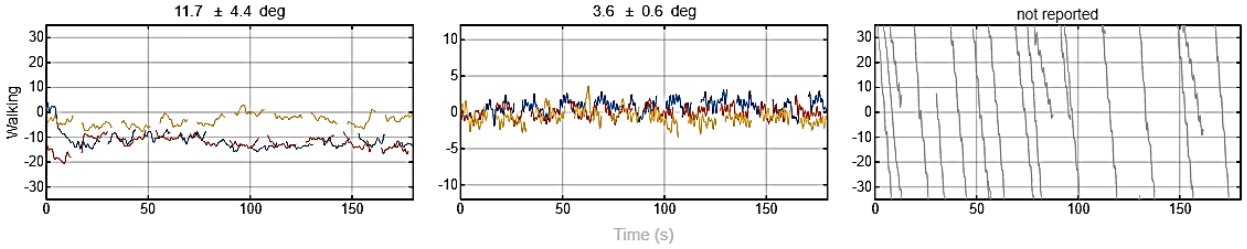


Figure 2: Head-angle traces for all (3 trials each of 5 tasks) at 180 s. Flexion-extension and lateral bending use each trial neutral reference. Axial rotation is linearly de-trended. (Walking axial rotation is shown only to identify turning and is not an outcome)

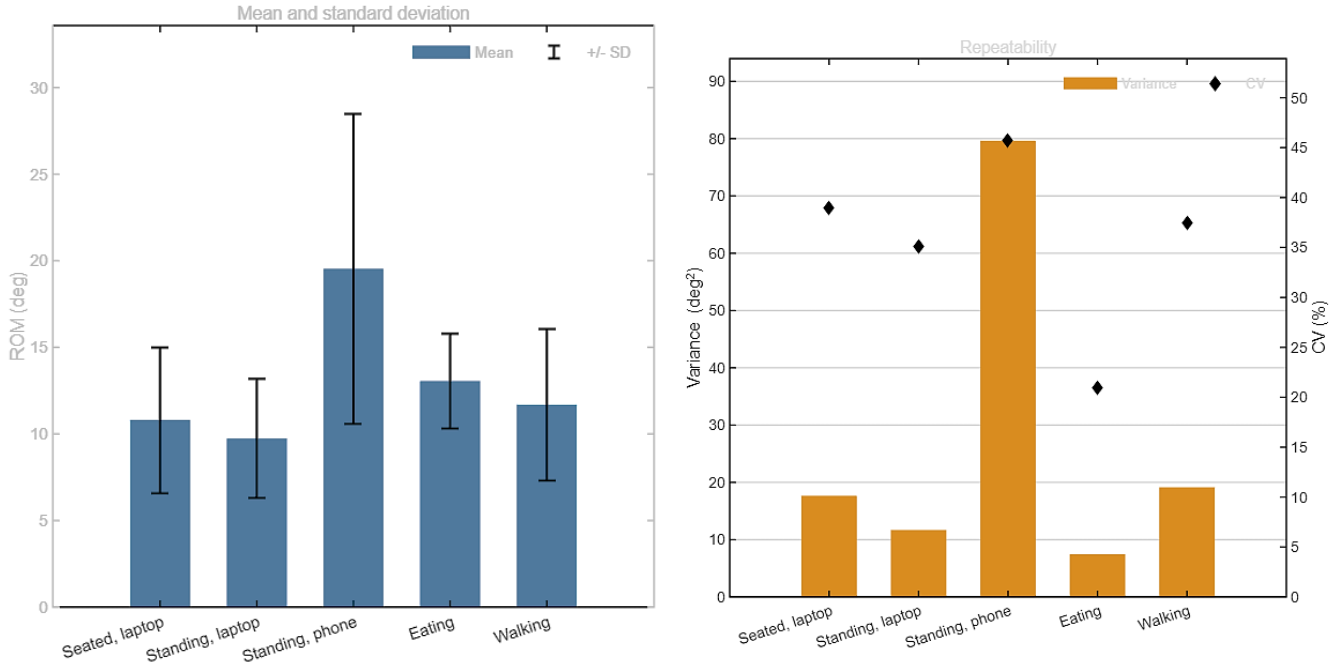


Figure 3: Flexion-extension mean and SD, with sample variance and CV across the three trial-level excursions for each task. Lower SD, variance, and CV indicate more similar repetitions within this dataset.

Discussion

The recordings produced task-specific movement patterns, but repeatability differed considerably. Seated laptop work had a lower absolute SD and CV than standing phone use. SD and variance describe absolute spread; CV describes spread relative to the task mean.

The 1.12 deg quiet-hold noise was small compared with the 2.7-8.9 deg between-trial SD. Neutral phone inclination spanned 20.5 deg across the fifteen recordings. The head frame is rebuilt for each trial from that trial neutral gravity direction, so this spread carries into every trial-level result. Mounting, reference posture, and natural task performance may have contributed to the differences between trials.

Walking required separate treatment because each lap ended in a whole-body turn. Removing these periods retained 77% of samples, but the remaining signals still include trunk sway and impact, so the walking results describe head movement during the retained straight segments only.

Future work should use a rigidly indexed forehead mount, add a trunk sensor, and validate the output against optical motion capture. More participants and trials are required before inferential statistics or clinical claims.

Limitations

One forehead sensor measured head orientation in space and could not separate cervical motion from trunk motion. One healthy participant and three trials per task cannot support generalization to a population or a clinical group. Axial rotation depended on gyroscope integration and drift correction. No optical motion-capture system or clinical cervical-range instrument was available to establish measurement validity.

Conclusion

The study met its feasibility objective: fifteen recordings processed through one fixed workflow, giving task-level comparison of head excursion within this participant, but not validated cervical range-of-motion values.

References

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